
Module 1 and 2 assessment questions.

1. Consider a case, a folder has multiple files and how would copy it to destination machine path (Try using SCP, cp options in Linux)
2. Host a FTP and SFTP server and try PUT and GET operations.
3. Explore with Wireshark/TCP-dump/cisco packet tracer tools and learn about packets filters.
4. Understand linux utility commands like - ping, arp (Understand each params from ifconfig output)
5. Understand what happens when duplicate IPs configured in a network.
6. Understand how to access remote system using (VNC viewer, Anydesk, teamviewer and remote desktop connections)
7. How to check your default gateway is reachable or not and understand about default gateway.
8. Check iwconfig/ifconfig to understand in detail about network interfaces (check about interface speed, MTU and other parameters)
9. Log in to your home router's web interface (usually at 192.168.1.1 or 192.168.0.1) and check the connected devices list.
10. Explain how a DHCP server assigns IP addresses to devices in your network.
11. Using a terminal, connect to a remote machine via SSH and telnet.

1.

```
dharanish@DHARANISH-LOQ:/mnt/d/Solar Irradiance Forecasting$ cp -r test_plots "d"
dharanish@DHARANISH-LOQ:/mnt/d/Solar Irradiance Forecasting$ cp -r test_plots "/mnt/d/"
dharanish@DHARANISH-LOQ:/mnt/d/Solar Irradiance Forecasting$ scp -r test_plots "/mnt/d/"
dharanish@DHARANISH-LOQ:/mnt/d/Solar Irradiance Forecasting$ scp -r test_plots "/mnt/d/"
dharanish@DHARANISH-LOQ:/mnt/d/Solar Irradiance Forecasting$
```

2.

```

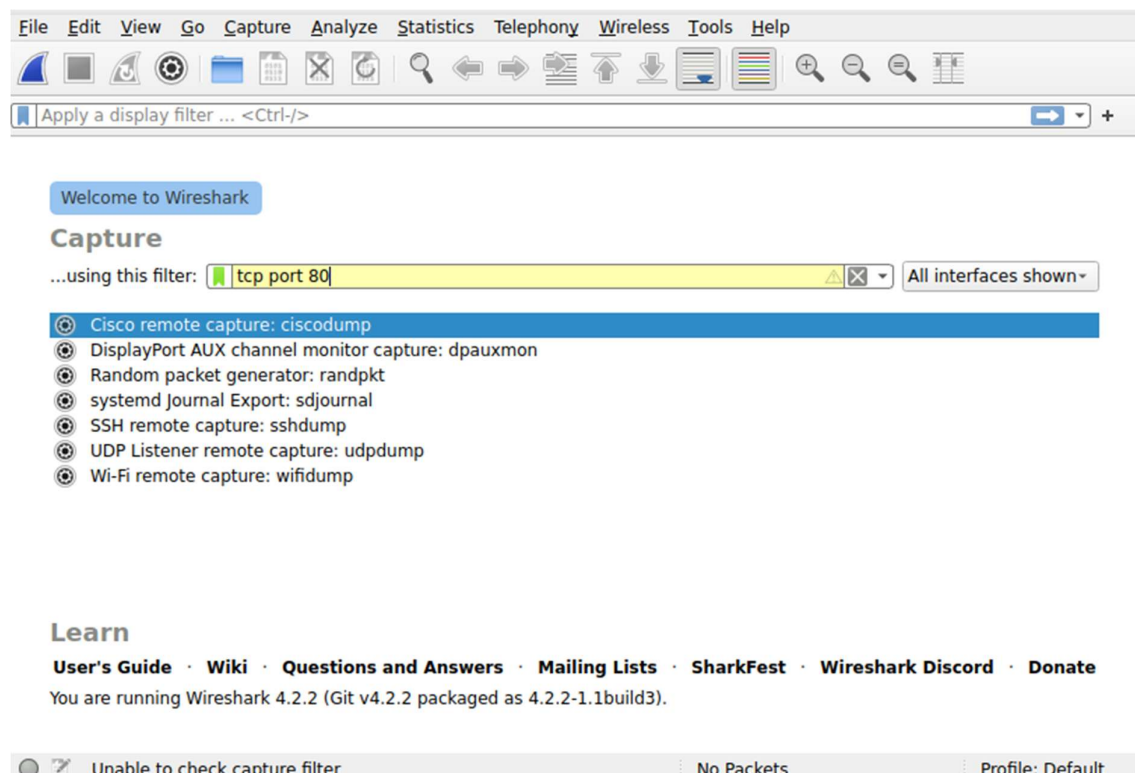
dharanish@DHARANISH-LOQ:~$ sudo systemctl restart vsftpd
[sudo] password for dharanish:
dharanish@DHARANISH-LOQ:~$ sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /lib/systemd/systemd-sysv-install
Executing: /lib/systemd/systemd-sysv-install enable vsftpd
dharanish@DHARANISH-LOQ:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/lib/systemd/system/vsftpd.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-03-27 06:13:13 IST; 10s ago
     Main PID: 1071 (vsftpd)
        Tasks: 1 (limit: 9339)
       Memory: 864.0K
          CPU: 9ms
      CGroup: /system.slice/vsftpd.service
              └─1071 /usr/sbin/vsftpd /etc/vsftpd.conf

Mar 27 06:13:13 DHARANISH-LOQ systemd[1]: Starting vsftpd FTP server...
Mar 27 06:13:13 DHARANISH-LOQ systemd[1]: Started vsftpd FTP server.

dharanish@DHARANISH-LOQ:~$ sudo nano /etc/vsftpd.conf
dharanish@DHARANISH-LOQ:~$ ftp localhost
Connected to localhost.
220 (vsFTPd 3.0.5)
Name (localhost:dharanish): dharanish
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> put test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||18224|)
550 Permission denied.
ftp> exit
221 Goodbye.

```

3.



```

dharanish@DHARANISH-LOQ:~$ sudo tcpdump -i eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
06:19:21.657024 IP 172.25.253.115.58373 > prod-ntp-4.ntp1.ps5.canonical.com.ntp: NTPv4, Client, length
48
06:19:21.838739 IP prod-ntp-4.ntp1.ps5.canonical.com.ntp > 172.25.253.115.58373: NTPv4, Server, length
48
06:19:27.756371 ARP, Request who-has 172.25.253.115 (00:15:5d:e9:95:12 (oui Unknown)) tell DHARANISH-L
OQ.mshome.net, length 28
06:19:27.756418 ARP, Reply 172.25.253.115 is-at 00:15:5d:e9:95:12 (oui Unknown), length 28
06:19:28.194080 ARP, Request who-has DHARANISH-LOQ.mshome.net tell 172.25.253.115, length 28
06:19:28.195229 ARP, Reply DHARANISH-LOQ.mshome.net is-at 00:15:5d:e9:92:49 (oui Unknown), length 28
^C
6 packets captured
6 packets received by filter
0 packets dropped by kernel

```

4.

```

dharanish@DHARANISH-LOQ:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=31.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=26.1 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=34.1 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=117 time=33.5 ms
^C
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 26.073/31.264/34.090/3.165 ms
dharanish@DHARANISH-LOQ:~$ arp -a
DHARANISH-LOQ.mshome.net (172.25.240.1) at 00:15:5d:26:2d:4d [ether] on eth0
dharanish@DHARANISH-LOQ:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.25.253.115 netmask 255.255.240.0 broadcast 172.25.255.255
    inet6 fe80::215:5dff:fee9:988f prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:e9:98:8f txqueuelen 1000 (Ethernet)
    RX packets 535 bytes 473417 (473.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 310 bytes 66058 (66.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 36 bytes 3815 (3.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 36 bytes 3815 (3.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

5.

Packet loss

Ip to Mac mapping errors

6.

```
dharanish@DHARANISH-LOQ:~$ ssh 22bcs034@172.16.16.200
```

```
*****
```

```
Welcome to MEPCOLINUX
```

```
*****
```

```
Keep Your Password Secure
```

```
*****
```

```
22bcs034@172.16.16.200's password:
```

7.

```
dharanish@DHARANISH-LOQ:~$ ip route
default via 172.25.240.1 dev eth0 proto kernel
172.25.240.0/20 dev eth0 proto kernel scope link src 172.25.253.115
dharanish@DHARANISH-LOQ:~$
```

8.

```
dharanish@DHARANISH-LOQ:~$ iwconfig
lo          no wireless extensions.

eth0        no wireless extensions.

dharanish@DHARANISH-LOQ:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 172.25.253.115  netmask 255.255.240.0  broadcast 172.25.255.255
    inet6 fe80::215:5dff:fee9:988f  prefixlen 64  scopeid 0x20<link>
    ether 00:15:5d:e9:98:8f  txqueuelen 1000  (Ethernet)
    RX packets 656  bytes 611434 (611.4 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 384  bytes 72645 (72.6 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 42  bytes 4529 (4.5 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 42  bytes 4529 (4.5 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

dharanish@DHARANISH-LOQ:~$ ip addr any
Command "any" is unknown, try "ip address help".
dharanish@DHARANISH-LOQ:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:e9:98:8f brd ff:ff:ff:ff:ff:ff
    inet 172.25.253.115/20 brd 172.25.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fee9:988f/64 scope link
        valid_lft forever preferred_lft forever
dharanish@DHARANISH-LOQ:~$ |
```

9.

```
dharanish@DHARANISH-LOQ:~$ ip route
default via 172.25.240.1 dev eth0 proto kernel
172.25.240.0/20 dev eth0 proto kernel scope link src 172.25.253.115
http://172.25.240.1
```

10.

DORA Process

11.

ssh [22bcs034@172.16.16.200](ssh://22bcs034@172.16.16.200)

telnet 172.16.16.200